High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance

Michael Cusack-Nelkin
mc5882@columbia.edu

Abstract

Each day, Nasdaq publishes auction imbalance information during the ten minutes leading up to the closing auction in order to attract liquidity and improve price discovery. In doing so it prints a mechanical signal that is learnable and strongly predictive of returns into the 16:00 close. We build a full trading system around this signal on the principles of classical statistical arbitrage: a forecast of 15:50 - 16:00 returns, a 10-factor covariance model, and a model of trading costs, assembled as the terms of a single-period maximization problem which is solved for an optimal portfolio of dollar holdings. Our features are time-series and cross-sectional transformations of signed dollar imbalance. We model return forecasts by boosting a lasso regression with xgboost, which robustly achieves a 29.6% average out-of-sample rank information coefficient through a rolling cross-validation. Backtested over 110 days, the strategy achieves a Sharpe ratio of 8.53 at a gross market value (GMV) of $91.3M. We argue that this elevated performance is expected given a high-IC forecast over a ten-minute horizon, but also that there are tight limits on GMV capacity and latency, with diminishing marginal PnL above ~$90M in GMV and one third of the forecasted return realized within the first one hundred microseconds after the imbalance is disseminated. Given these constraints, we believe that this strategy would see the most success if integrated into an optimal execution pipeline serving a larger statistical arbitrage portfolio, allowing for consistent improvement while rebalancing large positions near the close.

1 Motivation and Main Ideas

The Nasdaq equities closing auction occurs at 16:00:00 each trading day, and relevant auction information is published at a regular interval from 15:50:00 onward (Nasdaq Stock Market LLC 2024). At about 15:50:00, the exchange begins publishing information on the imbalance of auction orders. That is, the number of shares from on-close orders that will be left over after matching the maximum number of shares at the calculated auction execution price, as well as the side (buy/sell/none) of the leftover shares. The purpose of the auction imbalance publication is to attract auction liquidity to the opposite side of the imbalance, maximizing the number of shares executed and providing the best possible end-of-day price discovery. Therefore, a thesis exists: auction imbalance published from 15:50:00 through the close provides a mechanical signal that is learnable and strongly predictive of returns over that horizon. We will refer throughout to *dollar imbalance*, calculated as the number of imbalance shares multiplied by the exchange-estimated auction execution price at 15:50:00, signed by the side of the imbalance shares. Dollar imbalance and its time-series and cross-sectional transformations will be the basis of our return forecasts throughout.

2 Methodological Choices and Innovations

This trading system is designed using optimal portfolio construction, the method underpinning classical statistical arbitrage. Given a single "period" or "horizon", we forecast per-asset returns, estimate their variance and covariance, and estimate the costs of trading. We then assemble these pieces as the terms of a maximization problem and solve for the optimal portfolio of dollar holdings

$p^* \in \mathbb{R}^n$ over our horizon. The reason we use this framework is that it allows for non-trivial risk and cost dynamics to be accounted for in maximizing PnL over a given time horizon, such as factor-neutrality and hedging, lasso-like cost penalties, and instantaneous impact modeling. This problem is also at-worst SOCP and we make use of free-access solvers such as Clarabel via CVXPY (Diamond and Boyd 2016). The decision to use a single horizon, rather than multiple, is discussed in Section A.2.

$$\begin{aligned} \max_{p} \quad & \underbrace{p^T f}_{\text{alpha}} - \underbrace{\frac{\lambda}{2} p^T \Sigma_r p}_{\text{risk}} - \underbrace{C(u)}_{\text{spread}} - \underbrace{I(u)}_{\text{impact}} \\ \text{s.t.} \quad & p^T L_{mkt} = 0 \\ & \sqrt{p^T \Sigma_r p} \leq \sigma_{\$} \end{aligned} \tag{1}$$

Equation 1 is the maximization problem we solve each day, where $f \in \mathbb{R}^n$ is a vector of return forecasts, $u = p - p_0, u \in \mathbb{R}^n$ is the vector of trades to be made over the horizon, $\Sigma_r \in \mathbb{R}^{n \times n}$ is a covariance matrix of returns, $C(u)$ is the cost of trading, and $I(u)$ is the instantaneous price impact of our trades. We discuss each term of the maximization problem, as well as the constraints, in detail in Section 2.1 through Section 2.4.

2.1 Forecasting

2.1.1 Features

The primary input for our trading system is a forecast of returns from 15:50 to the 16:00 closing auction. Making use of the strength of this signal, we take a keep-it-simple approach to modeling. We use two core features, and each is a transformation of the dollar imbalance, signed by the side of the imbalance. One is a time-series feature, which provides information across days, and one is cross-sectional, giving us information across assets, within days.

First, we call our time-series feature the "signed imbalance ratio". This is simply the signed dollar imbalance divided by an exponentially weighted moving average of past daily dollar volume with decay $\alpha = 0.20$. Second, we divide the signed dollar imbalance by auction participation (the total liquidity available as limit-on-close orders as of 15:50), and rank the result across stocks each day. This ranking is then normalized using Equation 2 to arrive at our "normalized rank imbalance" feature which is constrained to the range $[-1, 1]$.

$$\begin{aligned} & \boxed{\text{signed imb. ratio} = \frac{\text{signed dollar imbalance}}{\text{EWMA(dollar volume)}}} \\ & \text{imb. part.} = \frac{\text{signed dollar imbalance}}{\text{total auction dollars}} \\ \Rightarrow & \boxed{\text{norm. rank imb.} = 2\,\frac{\text{rank(imb. part.)} - \text{min(rank)}}{\text{max(rank)} - \text{min(rank)}} - 1} \end{aligned} \tag{2}$$

The combination of these features gives us a strong indication of the assets most likely to move each day. We also use the cross-sectionally demeaned return from 15:45 to 15:50 ("demeaned preemptive return") as a signal of how much of the imbalance-driven return may have been realized just before the close, which could possibly lower the forecast magnitude.

2.1.2 Models

The best forecasting model we found was a lasso regression combined with xgboost. We use this combination because our most powerful feature, the normalized rank imbalance, is inherently linear; however, the other features introduce weaker but significant non-linearity.

Using our two core features, as well as the demeaned preemptive return and interaction terms, we train a lasso model (Tibshirani 1996) with $\alpha = 10^{-4}$. We then train an xgboost model (Chen and Guestrin 2016) on the same features against the lasso regression's residuals (hyperparameters in Table 1) and combine the two by a simple sum (Equation 3). Using time-series cross-validation, this model achieves an average IC across folds of 29.6%, with a standard deviation of 2.86%, and out-of-sample t-statistic of $t = 2.99$.

The complexity of these feature transformations and the combination of two models may seem like overkill for a powerful short horizon signal such as signed dollar imbalance. However, we found that our method improves average IC from 24.5% to 29.6% and reduces the standard deviation of IC from 5.77% to 2.86% when compared with a simple z-scored imbalance and univariate OLS under out-of-sample rolling cross-validation. We also achieve near-homoskedasticity and near-normally distributed residuals as shown in Figure 1, a good indication of predictive power without overfitting.

$$\begin{aligned} \hat{\beta} &= \arg\min_{\beta} \quad \frac{1}{2}|y - X\beta|_2^2 + \alpha|\beta|_1 \quad \text{(lasso problem)} \\ \hat{f} &= X\hat{\beta} + \text{xgboost}(X) \end{aligned} \tag{3}$$

Table 1: XGBoost Parameters

Parameter	Value
n_estimators	50
max_depth	4
learning_rate	0.01
L1 reg.	10^{-6}
L2 reg.	10^{-10}
objective	squared error loss
random_state	29385

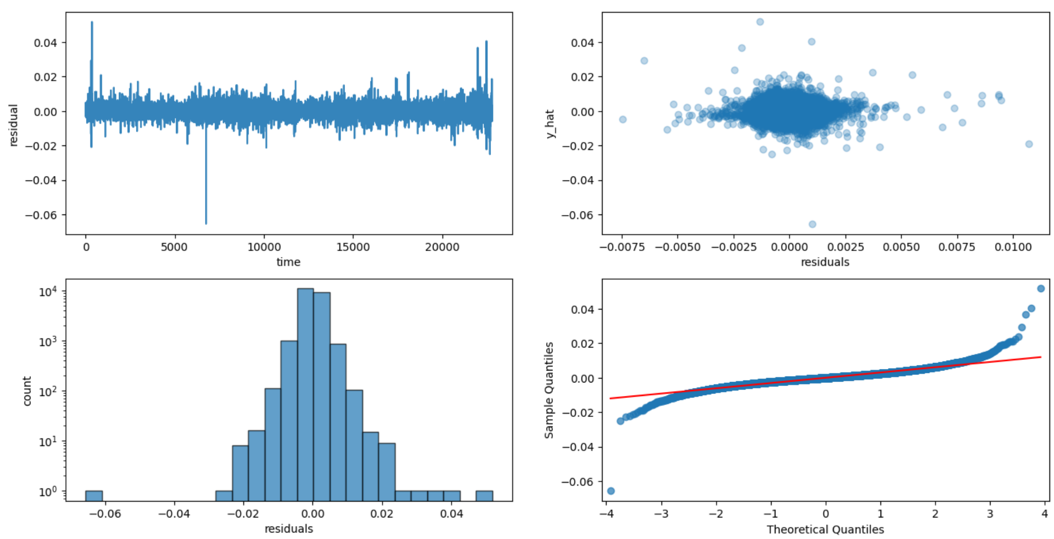

Figure 1: Residual Diagnostics

2.2 Covariance Estimation

We represent Σ_r using a factor model of standardized returns. We first volatility-adjust the $T \times n$ returns matrix X into $\tilde{X}$ using a causal EWMA volatility estimate (lagged one day for causality), and take the best rank-10 singular value decomposition $\tilde{X} \approx U_{10}S_{10}V_{10}^T$. Because $\tilde{X}$ is unitless, the resulting loadings and residual variances must be mapped back into return units before the optimizer can use them. We do this with $C = \mathrm{Diag}(\mathrm{std}(X))$, the diagonal matrix of per-asset sample return volatilities, at the final computation of Σ_r using the unitless $\tilde{\Sigma}$ so that Σ_r carries units of squared daily return. The full model is estimated on daily bars over 2023-06-01 to 2025-06-01, a window deliberately disjoint from the 2025-06-11 to 2026-06-11 period over which the forecast and backtest run, so the covariance is used strictly out of sample.

$$
\begin{aligned}
&C = \mathrm{Diag}(\mathrm{std}(X)) \in \mathbb{R}^{n\times n} \qquad F = U_{10} \\
&L = S_{10}\, V_{10}^T \qquad D = \mathrm{Diag}(\mathrm{var}(\tilde{X} - F\, L)), \quad D \in \mathbb{R}^{n\times n} \\
&\Rightarrow \Sigma_F = \frac{1}{T-1} F^T\, F \\
&\Rightarrow \tilde{\Sigma} = L^T\, \Sigma_F\, L + D \\
&\boxed{\Sigma_r = C^T\, \tilde{\Sigma}\, C}
\end{aligned}
\tag{4}
$$

Compared with the sample covariance matrix of returns Σ, using this factor model of returns increases accuracy and drastically reduces numerical issues in solving for p^*. Empirically, we have found that the condition number of sample covariance is on the order of 10^5 and it is not positive semi-definite (PSD), while the factor-modeled Σ_r has a condition number on the order of 10^2, and is PSD. It also allows us to use factor-neutrality constraints in the optimizer, discussed further in Section 2.4, which prevents accidental concentration of risk along any particular direction of return variance. Without this model, the optimizer would run into numerical issues solving, and the integrity of the solve would be questionable at best due to very small eigenvalues leading to unrealistic near-zero risk directions that the optimizer would leverage heavily.

2.3 Price Impact Modeling

We model price impact as a volatility-adjusted square-root impact model (Tóth et al. 2011), as in Equation 5. Rather than fitting values for the impact parameter γ per asset, we experimented with multiple scalar values from the range $[0, 1]$, and decided on $\gamma \approx 0.15$, estimated by (Almgren et al. 2005) using institutional trading data from Citigroup. Since we are working on a ten-minute horizon with one round-trip trade that is latency constrained, we assume that the only impact cost is instantaneous or temporary impact (Almgren and Chriss 2001).

$$
\underbrace{I}_{\text{return impact}} = \gamma\, \sigma \sqrt{\frac{u}{\mathrm{ADV}}} \tag{5}
$$

where $\sigma \in \mathbb{R}^n$ is a vector of asset volatilities, $u = |p - p_0| \in \mathbb{R}^n$ are the absolute value of trades, and $\mathrm{ADV} \in \mathbb{R}^n$ is an EWMA of daily dollar volume.

2.4 Optimizer

As mentioned in Section 2, we make use of classical statistical arbitrage machinery to estimate the optimal dollar holdings over our trading horizon. The full maximization problem is stated in Equation 6. We use a single-period MVO along with spread cost and instantaneous impact terms, and use CVXPY (Diamond and Boyd 2016) to independently solve for the optimal dollar portfolio over each period. At each period we solve Equation 6. Note that $u = p$ since each portfolio is initially $p_0 = 0$.

$$
\begin{aligned}
\max_{p} \quad & p^T\, f - \frac{\lambda}{2} p^T\, \Sigma_r\, p - \frac{s}{2}|p| - \gamma\, \sigma\, \frac{|p|^{3/2}}{\sqrt{\mathrm{ADV}}} \\
\text{s.t.} \quad & p^T\, L_{mkt} = 0 \\
& \sqrt{p^T \Sigma_r p} \le \sigma_{\$}
\end{aligned}
\tag{6}
$$

We also subject our optimal portfolio to factor-neutral constraints, specifically neutrality against the factor of largest variance (the market factor) derived from the earlier SVD, and to expected dollar volatility constraints. This strategy is less sensitive to volatility than to costs and impact, since this is a relatively fast strategy with high feature ICs; however, the factor-neutrality guarantee enforces a reliance only on forecasted returns, not market movements generally. The expected dollar volatility ensures that the portfolio is never expected to move by more than some cap, which helps control risk and prevents the optimizer from suggesting a rapid increase in GMV.

3 Empirical Results

3.1 Backtest

We conduct our backtest over 110 days from [2026-01-01, 2026-06-11], and we attempt to get a best estimate of the Sharpe ratio of this strategy at its full-capacity GMV limit. Note from Section A.1 that the forecasting model was trained on [2025-06-11, 2025-12-31] and the covariance model on daily data from [2023-06-01, 2025-06-01]. The parameters for this run are in Table 3 and results are reported in Table 2 and Figure 3. Although such a high Sharpe ratio may be met with skepticism, the assumptions leading to such performance and the constraints inherent to it are discussed in Section 3.2. The length and number of assets included in the backtest were restricted by the retail cost of L3 and L2 data, discussed in Section A.1.

Table 2: Backtest Results

Result	Value
Mean GMV	\$69.3M
Max. GMV	\$91.3M
Sharpe	8.53
Gross PnL	\$6.4M
Spread Cost	\$1.8M
Impact Cost	\$1.7M
Net PnL	\$2.9M
Exp. Volatility	\$154K / day
Max. Drawdown	-\$91.9K

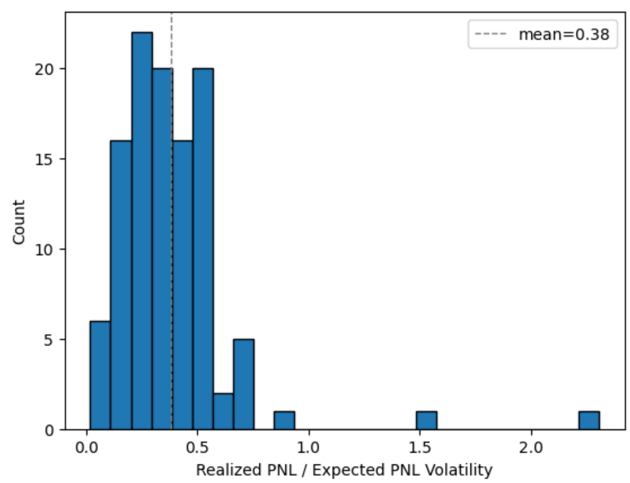

Figure 2: Absolute Gross PnL / Expected PnL Volatility

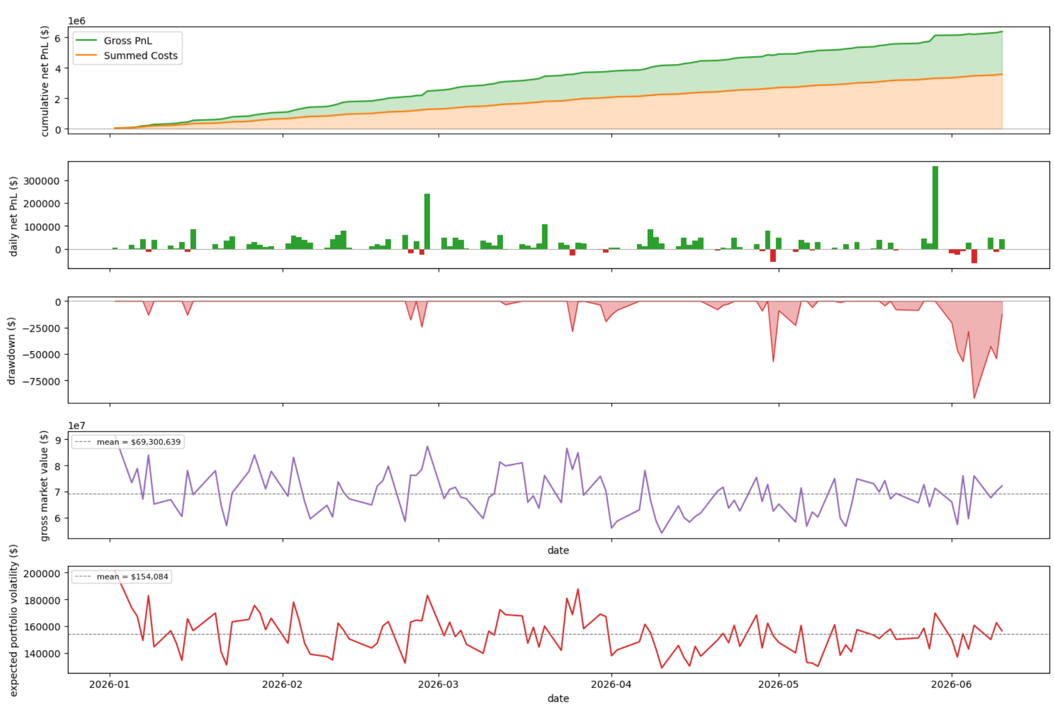

Figure 3: Backtest over 110 Days and 100 Assets

At these and similar parameter values, the optimizer produces relatively stable performance and gross market value (GMV). It has shallow drawdowns and a high Sharpe ratio, with small but consistent gains. Testing GMVs above this level leads to diminishing marginal PnL. Note also from Figure 2 that the mean absolute value of gross PnL is 38% of the expected PnL volatility, lower than the theoretical mean of $\sqrt{\frac{2}{\pi}} \approx 80\%$ from a half-normal distribution. This is the result of expected PnL volatility being calculated from daily return data, while the realized PnL is from a ten-minute holding period. Naively scaling the 80% expected mean daily volatility by clock-time, one would expect the realized volatility to be $\sqrt{\frac{10}{390}} \approx 16\%$ of daily volatility. But we observe $\frac{0.38}{0.80} \approx 48\%$, a much smaller reduction. This implies that 48% of daily volatility, or $0.48^2 \approx 23\%$ of daily variance, is realized during the ten-minute closing period, roughly 2.6% of the trading session. This is consistent with the "U-shaped" volatility curve empirically observed in microstructure studies (Bouchaud et al. 2018). It also means that the risk term in Equation 6 is over-penalizing by $\sim 4\times$ in units of variance, which is evidence that this strategy is cost-constrained, as expected with such a short horizon.

3.2 Capacity and Execution Constraints

This strategy's performance in Section 3.1 may seem overstated or raise concerns of bias or error. However, this level of performance is to be expected in any strategy using high IC features over a short horizon. It is important to understand the conditions for such strategies to work, though. The crucial thing is that this alpha is subject to serious capacity and execution constraints.

Given the short timeframe of trading, the capacity of this alpha is constrained to GMVs below ~$90M. Since we are effectively dumping the entire capacity into and out of the market at one instant around 15:50 and the closing auction, we have high instantaneous impact. Also, the nature of our exit strategy being MOC orders which offset the auction imbalance means that our trading reduces the effect of the imbalance on price, so we are doubly constrained.

Given the high price impact, we may be inclined to spread out trades over the ten-minute window; however, two things prevent this. First, Nasdaq only accepts MOC orders until 15:55 (Nasdaq Stock Market LLC 2024), so the exit MOC orders must be placed before then, restricting us to five minutes of trading. More importantly though, Figure 4 shows about one third of the forecasted return from 15:50 - close is realized within the first $100\,\mu$s after Nasdaq first disseminates the auction imbalance, and two thirds within the first 5ms. This means that the strategy is stuck with high price impact and that orders must be able to reach the exchange within $100\,\mu$s or else the alpha is too small to justify costs. This also means that in practice, the staggered nature of live messages leads to the risk that early arrivals' forecasted returns become realized before the entire universe has reported feature values.

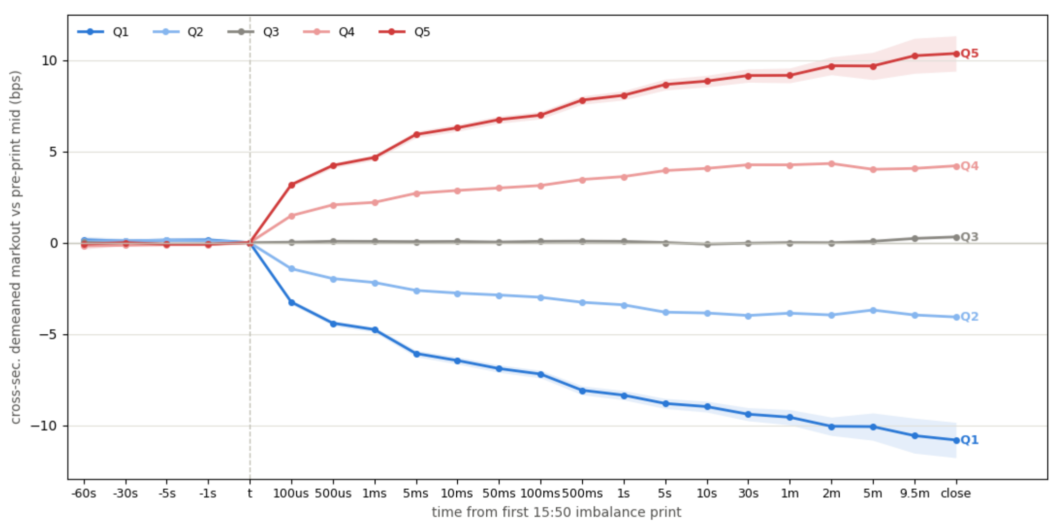

Figure 4: Returns Markout over Forecast Horizon Bucketed by Feature Quintiles

4 Future Work

The main areas for further research in this strategy are, as always, improving current forecasts and adding new ones. Most important would be adding a forecast of overnight returns. A likely first-pass improvement would be modeling the reversion of close auction returns into the next-day open. This would improve the strategy's capacity and PnL by lengthening the alpha's horizon and allowing for better management of price impact. This would also necessitate overnight volatility modeling.

As it stands, this system is a high-Sharpe, low-capacity opportunity. We believe that it would likely achieve the most success as part of an optimal execution pipeline within a larger, longer-horizon statistical arbitrage system. This way, the relatively small but consistent alpha demonstrated here can be captured while rebalancing larger positions near the closing auction.

A Appendix

A.1 Data

Throughout this work we use a combination of daily statistics, trades and BBO, and imbalance data. All data used here is sourced through Databento's XNAS.ITCH dataset (Databento, Inc. 2026). The exception to this is corporate-action adjustment factors, due to the expensive nature of such data, which are sourced via yfinance (Aroussi 2026) and sometimes inferred via Databento's daily data. Model fitting uses scikit-learn (Pedregosa et al. 2011) and xgboost (Chen and Guestrin 2016).

We chose to use one year of trades, BBO, and imbalance data from 2025-06-11 through 2026-06-11 in order to minimize expense and memory complexity. We split into train [2025-06-11, 2025-12-31] and test [2026-01-01, 2026-06-11] ranges and also use CAX-adjusted daily data from [2023-06-01, 2025-06-01] for the covariance modeling. The train/test split allowed us to avoid the difficult machinery of rolling-universe selection and instead use a single static universe of the 100 highest ADV Nasdaq equities from the train set. The reason we chose such a comparatively short backtest window and low number of assets was simply due to cost, as imbalance and trades with BBO data are level 3 and level 2 respectively, and are very expensive for an individual researcher.

A.2 Single vs Multi Period Optimization

The decision to use a "myopic" single-period optimizer instead of optimizing over a multi-period position path (Boyd et al. 2017) was driven by the nature of the alpha forecast we are using. The ten-minute horizon leaves little opportunity to trade and so frequent rebalancing costs more than our alpha forecast can overcome. If overnight forecasts are later developed (see Section 4) we would recommend moving toward multi-period planning to reduce price impact.

A.3 Hyperparameters

The hyperparameters used in Section 3.1 are listed below. The risk parameter λ is roughly on the inverse order of GMV ($\lambda = 10^{-6} \rightarrow$ GMV $\sim 10^6$). γ is the impact hyperparameter discussed in Section 2.3 and the volatility cap is discussed in Section 2.4.

Table 3: Backtest Parameters

Parameter	Value	Desc.
λ	10^{-6}	Risk param. on the order of ~ 1 / GMV
γ	0.15	Impact param. ranging from $[0, 1]$.
Volatility Cap	10^6	Expected portfolio PnL volatility cap ($).

References

Almgren, Robert, and Neil Chriss. 2001. "Optimal Execution of Portfolio Transactions." *Journal of Risk* 3 (2): 5–39. https://doi.org/10.21314/JOR.2001.041.

Almgren, Robert, Chee Thum, Emmanuel Hauptmann, and Hong Li. 2005. "Direct Estimation of Equity Market Impact." *Risk* 18 (7).

Aroussi, Ran. 2026. "Yfinance: Download Market Data from Yahoo! Finance's API." https://github.com/ranaroussi/yfinance.

Bouchaud, Jean-Philippe, Julius Bonart, Jonathan Donier, and Martin Gould. 2018. *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge, UK: Cambridge University Press.

Boyd, Stephen, Enzo Busseti, Steven Diamond, Ronald N. Kahn, Kwangmoo Koh, Peter Nystrup, and Jan Speth. 2017. "Multi-Period Trading via Convex Optimization." *Foundations and Trends in Optimization* 3 (1). https://doi.org/10.1561/2400000023.

Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–94. https://doi.org/10.1145/2939672.2939785.

Databento, Inc. 2026. "XNAS.ITCH: Nasdaq TotalView-ITCH Historical and Live Data." https://databento.com/datasets/XNAS.ITCH.

Diamond, Steven, and Stephen Boyd. 2016. "CVXPY: A Python-Embedded Modeling Language for Convex Optimization." *Journal of Machine Learning Research* 17 (83): 1–5.

Nasdaq Stock Market LLC. 2024. "Equity 4, Rule 4754: Nasdaq Closing Cross." Nasdaq Rulebook. https://listingcenter.nasdaq.com/rulebook/nasdaq/rules.

Pedregosa, F., G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, et al. 2011. "Scikit-Learn: Machine Learning in Python." *Journal of Machine Learning Research* 12: 2825–30.

Tibshirani, Robert. 1996. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society, Series B* 58 (1): 267–88.

Tóth, B., Y. Lempérière, C. Deremble, J. de Lataillade, J. Kockelkoren, and J.-P. Bouchaud. 2011. "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets." *Physical Review X* 1 (2): 021006. https://doi.org/10.1103/PhysRevX.1.021006.